Harikrishna Baskaran

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Summary

Software Engineer with 2 years of experience in software development, API integration, and system optimization. Skilled in Python, JavaScript, React.js, Node.js, FastAPI, MongoDB, PostgreSQL, Agentic AI, and Large Language Models (LLMs). Experienced in building web applications, AI-powered solutions, and improving operational efficiency through automation. Currently Pursuing an M.Sc. in Robotics and Autonomous Systems at the University of Lübeck. Seeking a Werkstudent position in Software Development, AI, or Digital Transformation to contribute expertise and innovative solutions.

Education

M.Sc. Robotics and Autonomous Systems

04/2026 – 04/2028 (Expected)

University of Lübeck - Lübeck, Germany

B.E. Computer Science and Engineering

S.A. Engineering College - Chennai, India 04/2024

Experience

Software Engineer

STW E-commerce Private Limited - Chennai, India 06/2024 - 02/2026

- Developed software applications to automate business processes and improve operational efficiency.
- Integrated REST APIs to streamline data synchronization and reduce repetitive manual tasks.
- Optimized database queries and application performance to improve processing efficiency.
- Collaborated with development teams to troubleshoot technical issues and maintain reliable applications.
- Maintained technical documentation to support development workflows and knowledge sharing.

Tech Stack: Python, SQL, MongoDB, Data Analysis, REST APIs, MS Excel, Technical Documentation

Trainee Software Engineer

STW E-commerce Private Limited - Chennai, India 03/2024 - 05/2024

- Developed CMS components to improve content management and streamline administrative workflows.
- Migrated web applications to improve performance, usability, and information accessibility.
- Maintained digital content, ensuring accurate information presentation and consistent documentation.
- Integrated APIs to streamline data workflows and improve information management.

Tech Stack: Vue.js, Node.js, JavaScript, MongoDB, REST APIs

Technical Skills

- **Programming & Frontend:** Python, JavaScript, React.js, HTML5, CSS3, SCSS, Bootstrap, Material UI, Chart.js
- **AI & Automation:** Agentic AI, Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), Tool Calling
- **Backend & API Development:** FastAPI, Node.js, REST APIs, GraphQL, API Integration, System Integration
- **Databases & Analytics:** MongoDB, PostgreSQL, SQL, Reporting Dashboards, Business Process Optimization
- **DevOps & Version Control:** Docker, CI/CD Pipelines, Git, GitHub, GitLab, Bitbucket
- **Tools & Collaboration:** Postman, MobaXterm, Context API, Microsoft Office, Technical Documentation, Cross-functional Collaboration

Projects

Medicinal Plant Identification Using Deep Learning

- Developed an end-to-end medicinal plant identification system, covering data preprocessing, model development, and deployment using deep learning techniques.
- Integrated CNN and Random Forest models to improve classification performance through combined deep feature extraction and traditional machine learning methods.
- Achieved 96.5% accuracy on a Kaggle dataset while generating medicinal information outputs for identified plants.

Tech Stack: Python · OpenCV · Matplotlib · CNN

Enterprise Resource Planning (ERP) Application

- Developed digital modules to manage employee attendance, room availability, inventory, and reporting workflows.
- Integrated biometric systems to automate attendance recording and real-time data synchronization.
- Built interactive dashboards to organize, visualize, and monitor employee attendance data.
- Implemented reporting tools to track operational information, sales transactions, and inventory availability.

Tech Stack: Python · React.js · MongoDB · REST APIs · Material UI

AI Support Desk with Agentic AI and RAG

- Integrated Qwen through Ollama for local LLM inference, automated ticket management, and response generation.
- Developed AI agents with tool calling, workflow automation, conversation memory, and human approval.
- Built RAG pipelines using document extraction, chunking, nomic-embed-text embeddings, MongoDB storage, and Qdrant vector indexing.
- Implemented hybrid retrieval using semantic search, BM25, reciprocal rank fusion, and source citations.

Tech Stack: Python · Ollama · Qwen3.5 · Qdrant · Agentic AI · RAG · Semantic Search · BM25

Languages

- English - C1 (CEFR)
- German - B1 (CEFR)